

HW00

Course: Description

January 01, 2026 at 23:59:00

Instructor

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Problem 1

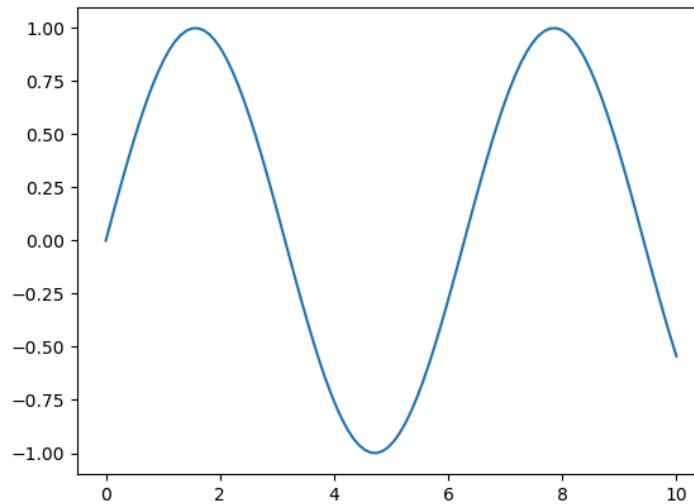
Parts (# pts)
a) Instructions
b) Instructions

a) Answer

b) Answer

Problem 2

Code (# pts)
Instructions

Figure 1: Sine Wave on $x = [0, 10]$

1	x:					
2	[0.	0.1010101	0.2020202	0.3030303	0.4040404	0.50505051
3	0.60606061	0.70707071	0.80808081	0.90909091	1.01010101	1.11111111
4	1.21212121	1.31313131	1.41414141	1.51515152	1.61616162	1.71717172
5	1.81818182	1.91919192	2.02020202	2.12121212	2.22222222	2.32323232
6	2.42424242	2.52525253	2.62626263	2.72727273	2.82828283	2.92929293
7	3.03030303	3.13131313	3.23232323	3.33333333	3.43434343	3.53535354
8	3.63636364	3.73737374	3.83838384	3.93939394	4.04040404	4.14141414
9	4.24242424	4.34343434	4.44444444	4.54545455	4.64646465	4.74747475
10	4.84848485	4.94949495	5.05050505	5.15151515	5.25252525	5.35353535
11	5.45454545	5.55555556	5.65656566	5.75757576	5.85858586	5.95959596
12	6.06060606	6.16161616	6.26262626	6.36363636	6.46464646	6.56565657
13	6.66666667	6.76767677	6.86868687	6.96969697	7.07070707	7.17171717
14	7.27272727	7.37373737	7.47474747	7.57575758	7.67676768	7.77777778
15	7.87878788	7.97979798	8.08080808	8.18181818	8.28282828	8.38383838
16	8.48484848	8.58585859	8.68686869	8.78787879	8.88888889	8.98989899
17	9.09090909	9.19191919	9.29292929	9.39393939	9.49494949	9.5959596
18	9.6969697	9.7979798	9.8989899	10.		]
19						
20	y:					
21	[0.	0.10083842	0.20064886	0.2984138	0.39313661	0.48385164
22	0.56963411	0.64960951	0.72296256	0.78894546	0.84688556	0.8961922
23	0.93636273	0.96698762	0.98775469	0.99845223	0.99897117	0.98930624

24	0.96955595	0.93992165	0.90070545	0.85230712	0.79522006	0.73002623
25	0.65739025	0.57805259	0.49282204	0.40256749	0.30820902	0.21070855
26	0.11106004	0.01027934	-0.09060615	-0.19056796	-0.28858706	-0.38366419
27	-0.47483011	-0.56115544	-0.64176014	-0.7158225	-0.7825875	-0.84137452
28	-0.89158426	-0.93270486	-0.96431712	-0.98609877	-0.99782778	-0.99938456
29	-0.99075324	-0.97202182	-0.94338126	-0.90512352	-0.85763861	-0.80141062
30	-0.73701276	-0.66510151	-0.58640998	-0.50174037	-0.41195583	-0.31797166
31	-0.22074597	-0.12126992	-0.0205576	0.0803643	0.18046693	0.27872982
32	0.37415123	0.46575841	0.55261747	0.63384295	0.7086068	0.77614685
33	0.83577457	0.8868821	0.92894843	0.96154471	0.98433866	0.99709789
34	0.99969234	0.99209556	0.97438499	0.94674118	0.90944594	0.86287948
35	0.8075165	0.74392141	0.6727425	0.59470541	0.51060568	0.42130064
36	0.32770071	0.23076008	0.13146699	0.03083368	-0.07011396	-0.17034683
37	-0.26884313	-0.36459873	-0.45663749	-0.54402111]		

See the Python code for this problem.

Code

```
./src/index.py Python
1  import numpy as np
2  import matplotlib.pyplot as plt
3  import os
4
5
6  def heading(problem):
7      print(f"\n-----\n {problem}\n-----")
8
9
10 def main():
11     #####
12     # p02 #
13     #####
14     heading("p02")
15
16     x = np.linspace(0, 10, 100)
17     y = np.sin(x)
18
19     print(f"x:\n{x}\nny:\n{y}")
20     with open("./outputs/text/s02.txt", "w", encoding="utf-8") as f:
21         f.write(f"x:\n{x}\nny:\n{y}")
22
23     plt.figure()
24     plt.plot(x, y)
25
26     save_path = os.path.join(os.path.dirname(__file__), "./outputs/figures/s02.png")
27     plt.savefig(save_path)
28     plt.show()
29
30 if __name__ == "__main__":
31     main()
```